

Determinants of Child Enrollment using ML

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Motivation

Despite decades of interventions, **Pakistan still faces critical child enrollment gaps**. Traditional statistical models fail to capture the **complex, non-linear realities** of poverty and education decisions.

This study applies **machine learning** to reveal hidden thresholds and interactions, transforming how we identify at-risk children and design targeted, effective policies beyond simplistic linear approaches.

Problem Statement

Traditional statistical models miss critical **non-linear patterns** in enrollment determinants. We need methods that capture **real-world complexity**—threshold effects, interactions, and predictive hierarchies—to effectively identify and support at-risk children.

Literature Review

Existing studies identify linear determinants—poverty, gender, distance—but fail to capture **complex interactions and threshold effects**.

Human capital theory and sociological frameworks explain individual barriers, yet cannot model their **combined real-world impact** or identify where interventions become most effective, creating a **critical methodological gap** for targeted policy design.

Data & Methods

We have used **MICS 2018-19 (Round 6)** data with about **65 thousand rows** compared econometric models (Logit, Ordered Logit) with **machine learning** (Random Forest, XGBoost, LightGBM) using 70/30 train/test split. Models are evaluated via ROC-AUC and minority class recall, with class weighting to address imbalance. Feature importance, non-linear splits, and threshold detection reveal how ML better captures real-world complexity than traditional linear assumptions.

Results & Model Performance

ML models outperform traditional methods significantly. XGBoost/LightGBM achieve 0.90 AUC vs. Logistic Regression's 0.86. Critical finding: **ML identifies 51% of non-enrolled children**, a 228% improvement. **Top determinants are child education, age, and wealth**. Wealth shows a **poverty trap below -0.8 index**. ML enables precise, threshold-based policy targeting for maximum impact.

Feature Engineering

Feature engineering transformed raw survey data into predictive variables. Categorical features (e.g., mother's education) were **one-hot encoded**. Continuous features (wealth index, age) were **standardized**. Key **binary** assets (electricity, bank account) were **kept intact**. Feature selection identified **25 top predictors from 60+ original variables**, reducing noise and focusing on policy-relevant determinants for machine learning analysis.

Model Evaluation

- Target the Poorest 20%:** Focus conditional cash transfers and enrollment
- Intervene Before Age 12:** Launch pre-emptive support programs
- Bundle Economic & Educational Support:** Combine interventions
- Adopt ML for District-Level Targeting:** Use predictive models to allocate limited resources to highest-risk sub-groups

Policy Recommendations

Model evaluation compared ML and econometric approaches across multiple metrics: **ROC-AUC, precision, recall**, and **F1-score**. Primary focus was on identifying non-enrolled children (Class 0 recall). **Tuned Weighted XGBoost** achieved the **highest AUC (0.888)** and a **51.4% recall for the minority class**, a **228% improvement** over baseline logistic regression. Cross-validation and learning curves confirmed robustness and minimal overfitting. Results were validated with **30% holdout** test data for reliability.

Limitations & Future Work

- Data Constraints:** MICS 2018 is cross-sectional; longitudinal data would better capture "dropout" vs. "never enrolled" dynamics.
- Qualitative Gaps:** ML cannot capture cultural nuances or "fear of safety" for female students, which require qualitative integration.

Figure # 1, Model Performances

Model Name	ROC-AUC	Accuracy	Class 0 Recall
Tuned Weighted XGBoost	0.8882	0.7529	0.5136
XGBoost (Unweighted)	0.8880	0.7752	0.3810
LightGBM (Best Overall)	0.8863	0.7725	0.3980
Weighted XGBoost	0.8828	0.7468	0.4524
Random Forest	0.8807	0.7515	0.2313
Decision Tree	0.8775	0.7454	0.3265
Logistic Regression (Baseline)	0.8496	0.7502	0.1565

Figure # 2, Dataset and Feature Details

Dataset Characteristic	Value / Count
Original Features	62
Original Observations	67,532
Categorical Features	7
Numerical Features	18
Used Features	25

Figure # 3, Confusion Matrix by Enrollment Category

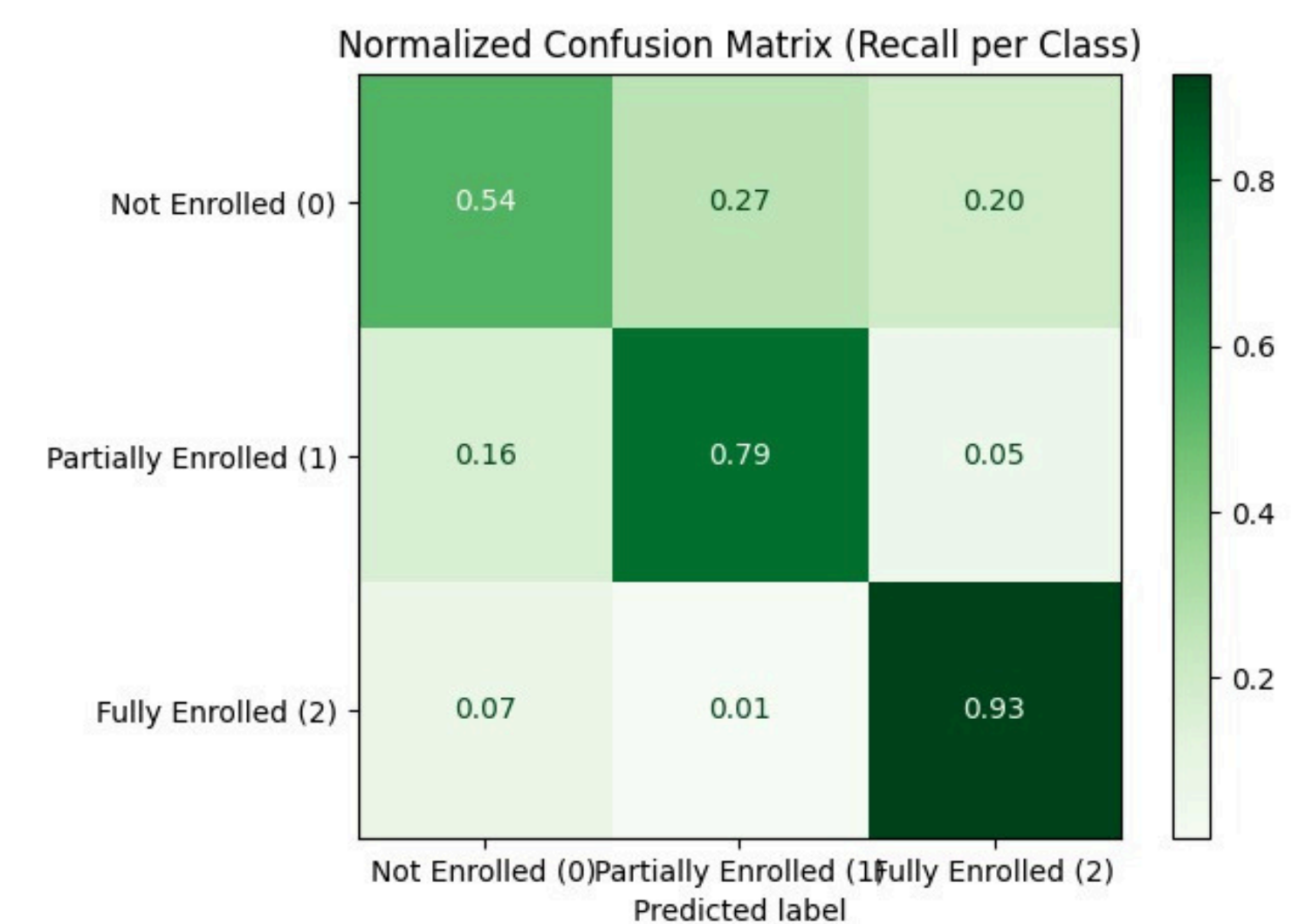


Figure # 4, ROC Curves - Model's measure of Class Diistribution

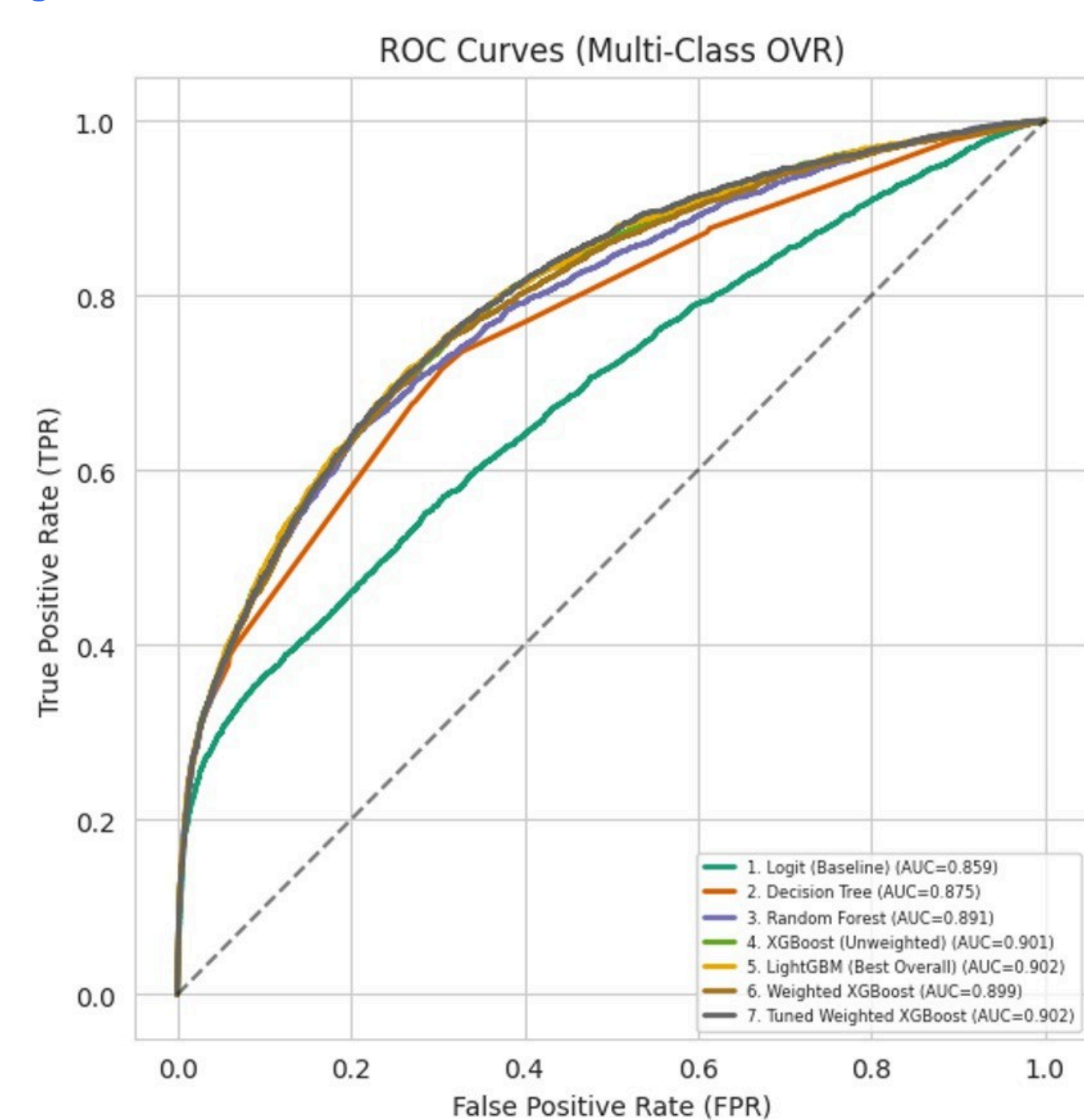


Figure # 5, Enrollment according to Class Distribution

